Why quantify urban segregation?

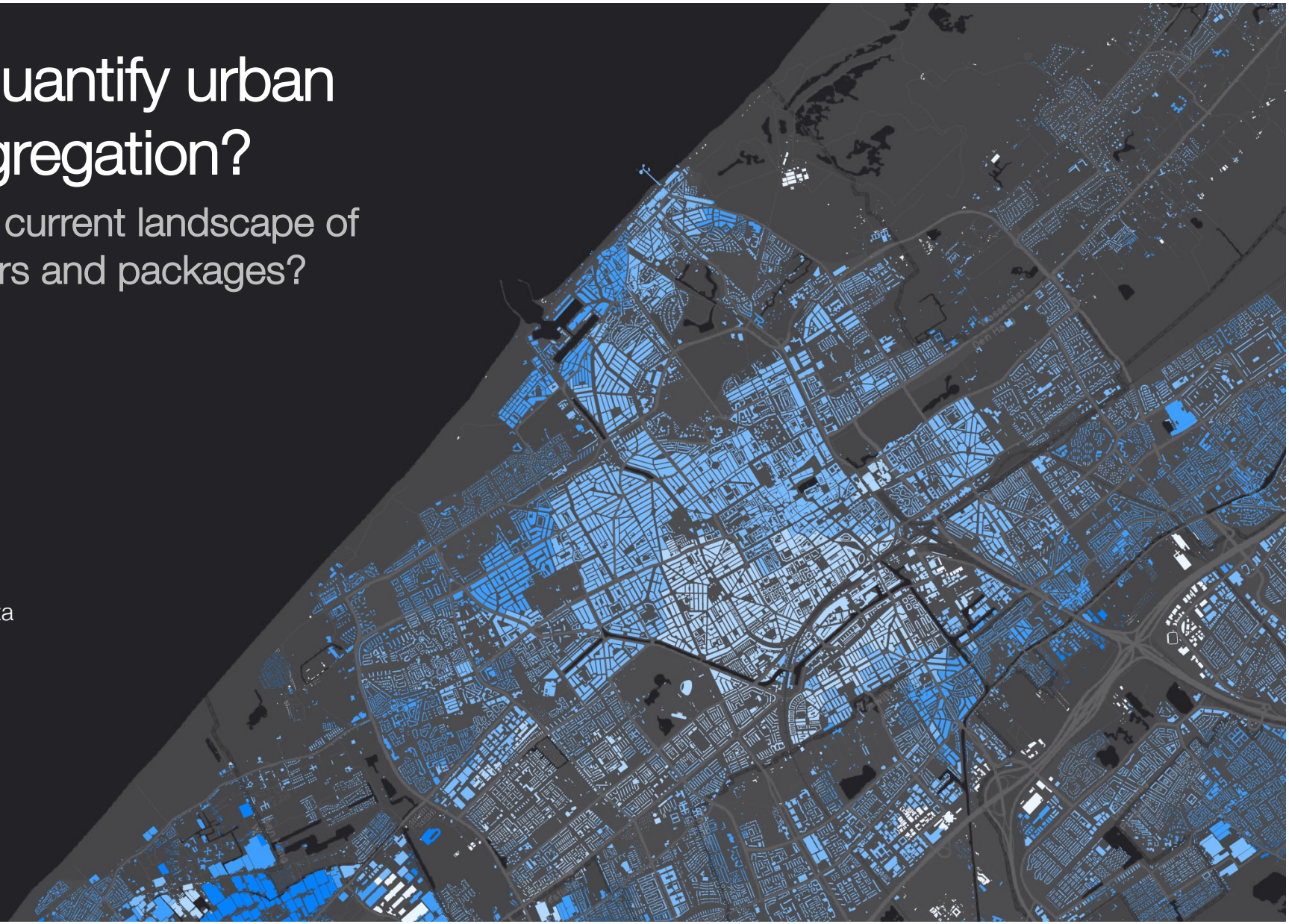
What's the current landscape of indicators and packages?

Ana Petrović
Clémentine Cottineau-Mugadza
Javier San Millán

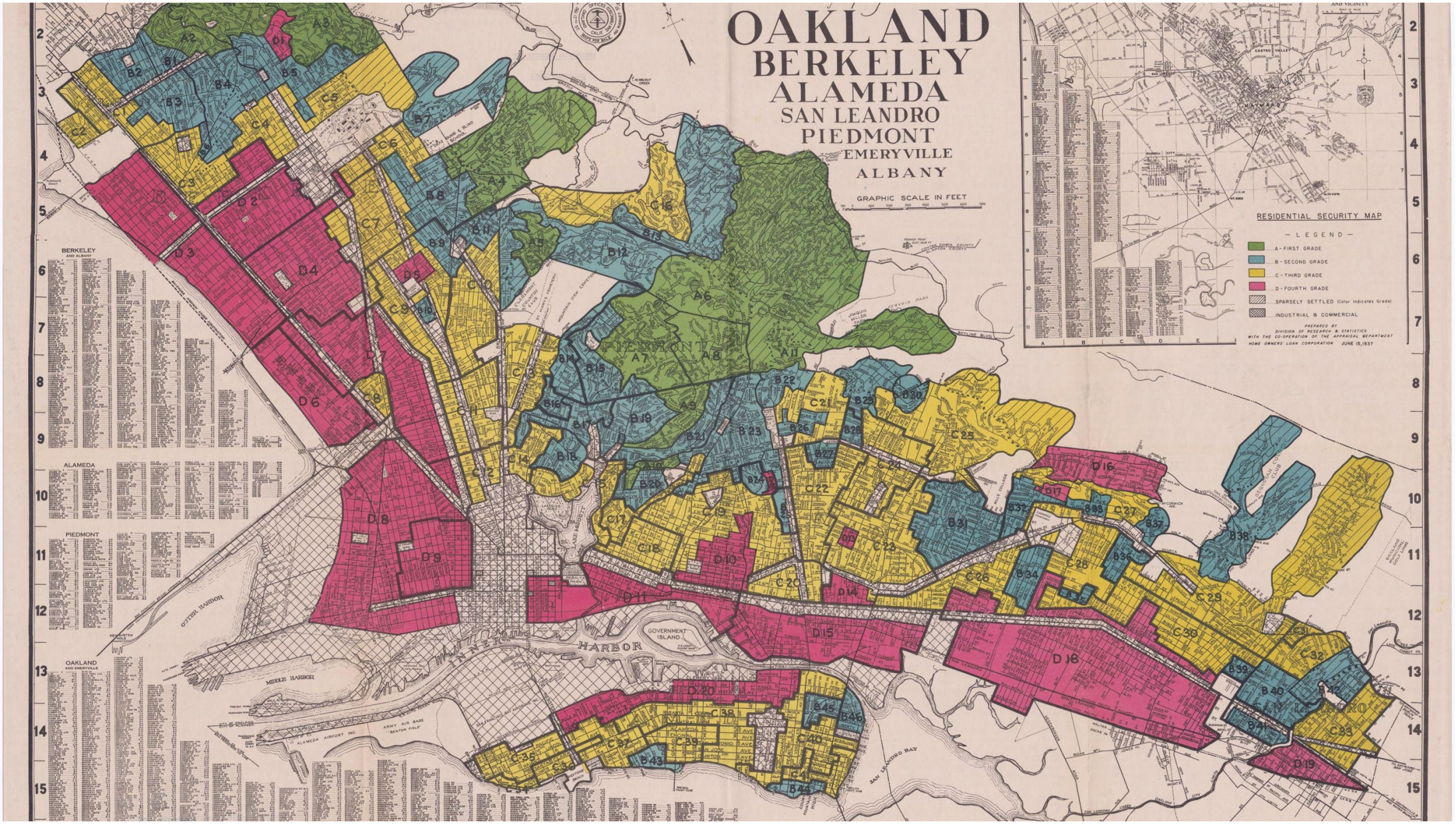
TU Delft

Rbanism

SEGUE






OAKLAND
BERKELEY
ALAMEDA
SAN LEANDRO
PIEDMONT
EMERYVILLE
ALBANY
GRAPHIC SCALE IN FEET
RESIDENTIAL SECURITY MAP
— LEGEND —
A - FIRST GRADE
B - SECOND GRADE
C - THIRD GRADE
D - FOURTH GRADE
SPARSELY SETTLED (Color Indicates Grade)
INDUSTRIAL & COMMERCIAL
PREPARED BY
DIVISION OF RESEARCH & STATISTICS
WITH THE CO-OPERATION OF THE APPRAISAL DEPARTMENT
HOME OWNERS' LOAN CORPORATION JUNE 15, 1937
BERKELEY AND ALBANY
ALAMEDA
PIEDMONT
OAKLAND AND EMERYVILLE
OTTER HARBOR
MIDDLE HARBOR
INNER HARBOR
GOVERNMENT ISLAND
SAN LEANDRO BAY
ALAMEDA AIRPORT INC.
ARMY AIR BASE "BENTON FIELD"
CASTRO VALLEY
HAYWARD

Why do we care?

- Segregation as the **spatial corollary of inequality**
- Segregation as a **mechanism of reproduction of inequalities** (e.g. neighbourhood effects)
- Segregation as a **threat to social cohesion**

But what is exactly segregation?

Defining segregation

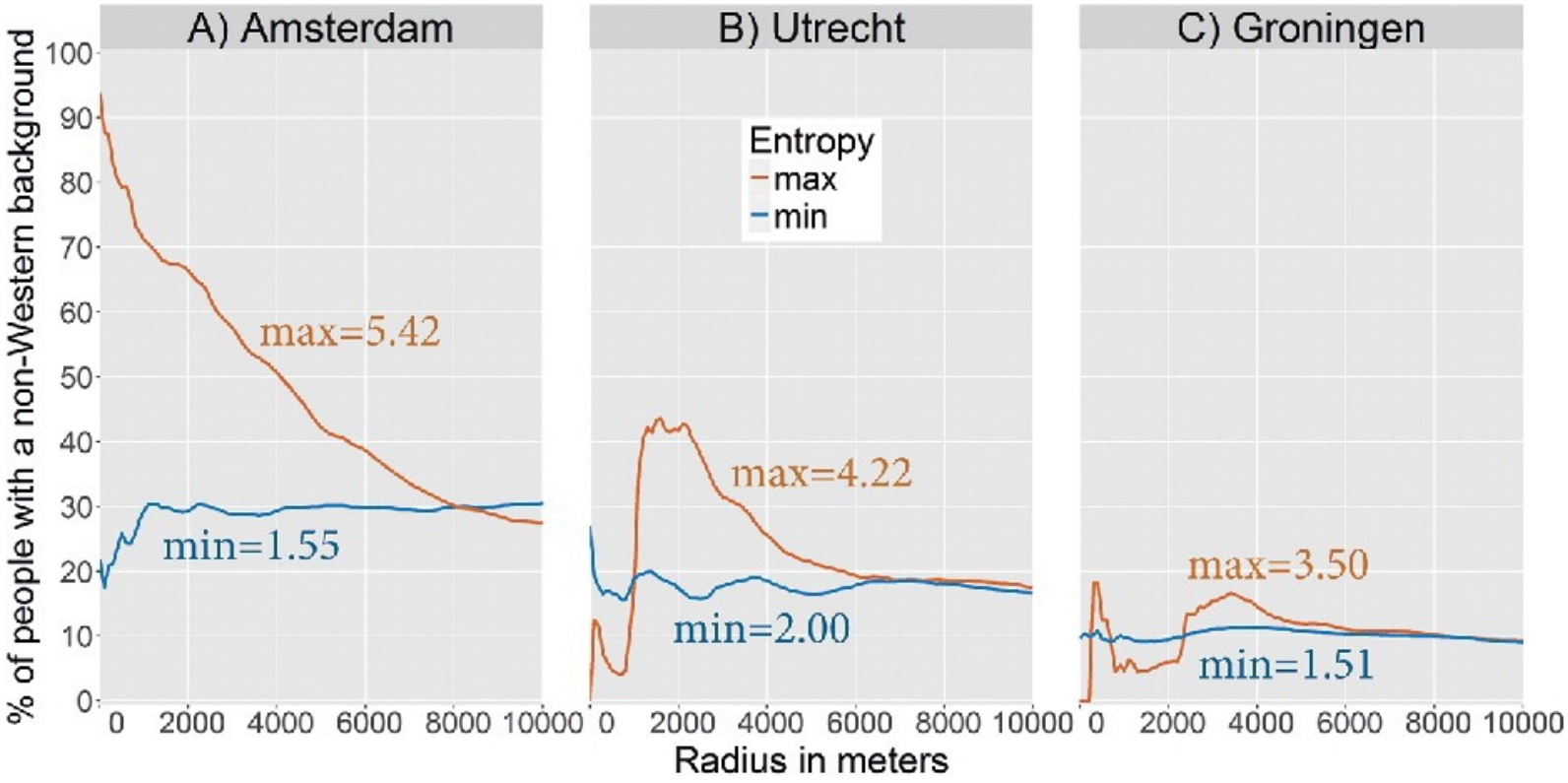
We propose a **brief, minimal definition**: segregation is the **separation of different social groups across space**

Agnostic definition with respect to the dimensions of segregation (e.g. Massey & Denton, 1988; Reardon & O'Sullivan, 2004) and to the type of social groups (i.e. racial, economic, religious, etc.)

Bonus debate: **can we apply this definition to non-spatial segregation in a strict sense** (e.g. occupational segregation)?

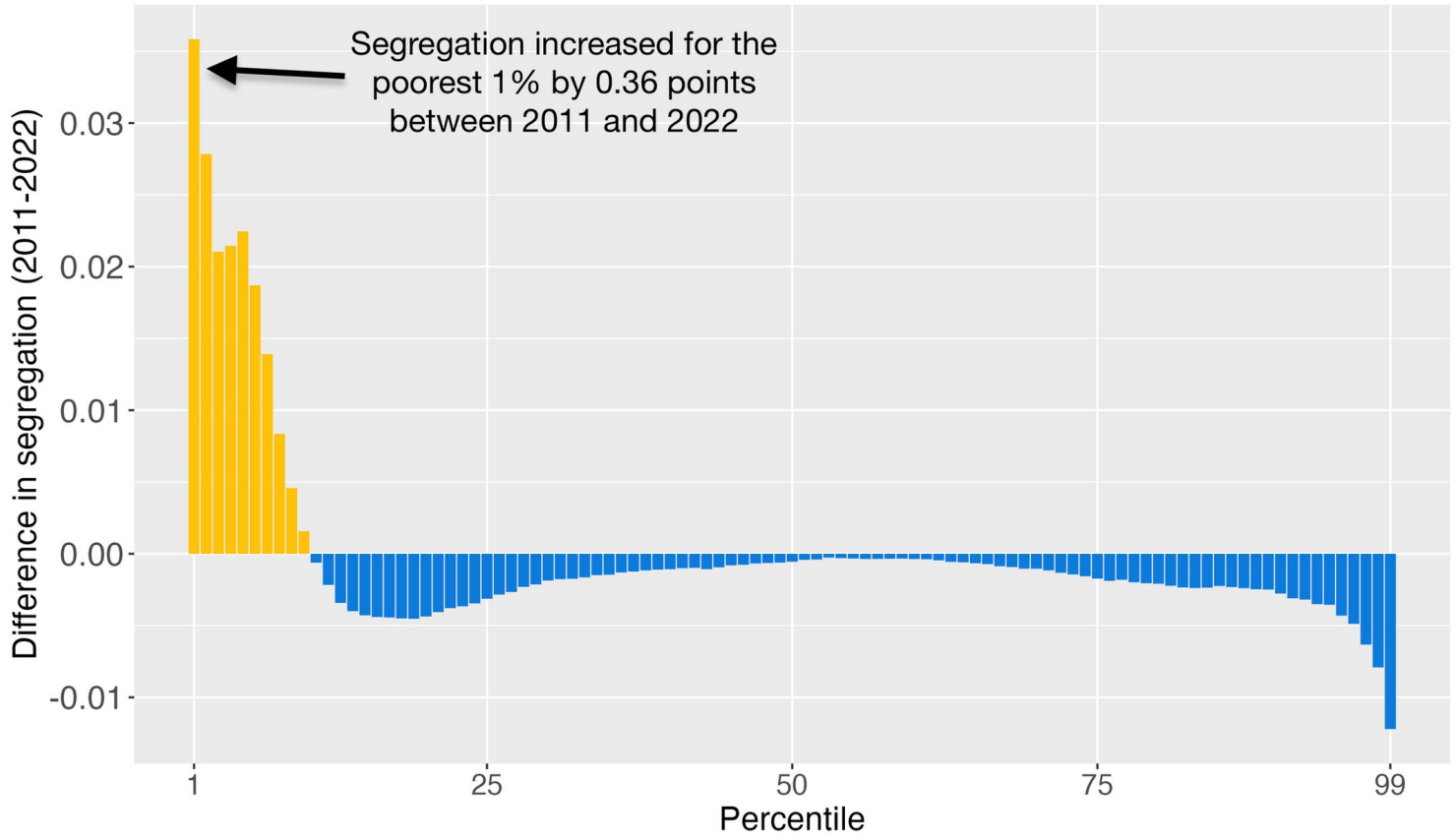
And why should we measure segregation?

Quantification allows us to compare, track and investigate



Extracted from: Petrović, A., Van Ham, M., & Manley, D. (2018). Multiscale measures of population: Within-and between-city variation in exposure to the sociospatial context. *Annals of the American Association of Geographers, 108*(4), 1057-1074.

Quantification allows us to compare, track and investigate



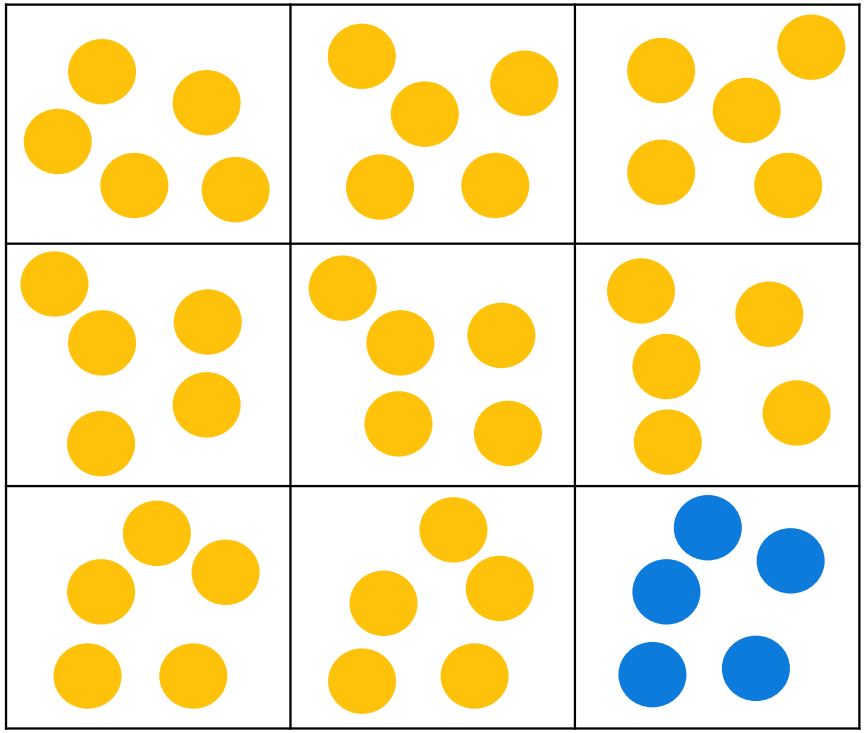
Extracted from: San Millán, J., Cottineau-Mugadza, C., & Van Ham, M. (2025). The economic urban divide: A detailed study of income inequality and segregation in Dutch urban areas (2011–2022). *Tijdschrift voor economische en sociale geografie, 116*(4), 508-530.

Is it easy to measure segregation?

Let's talk about problems

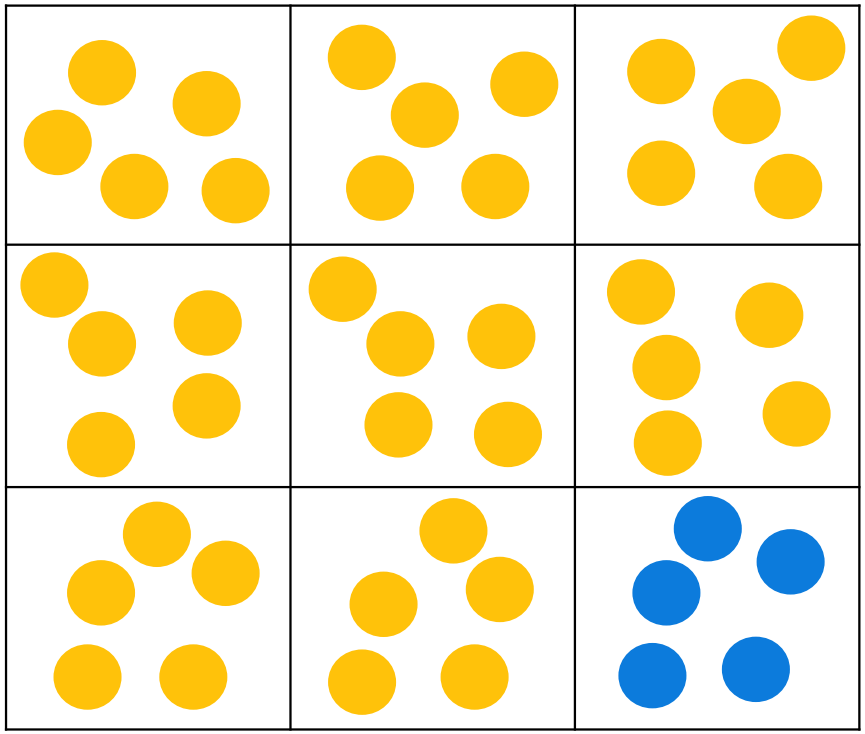
The problem of continuous variables

Categorical variable-based segregation

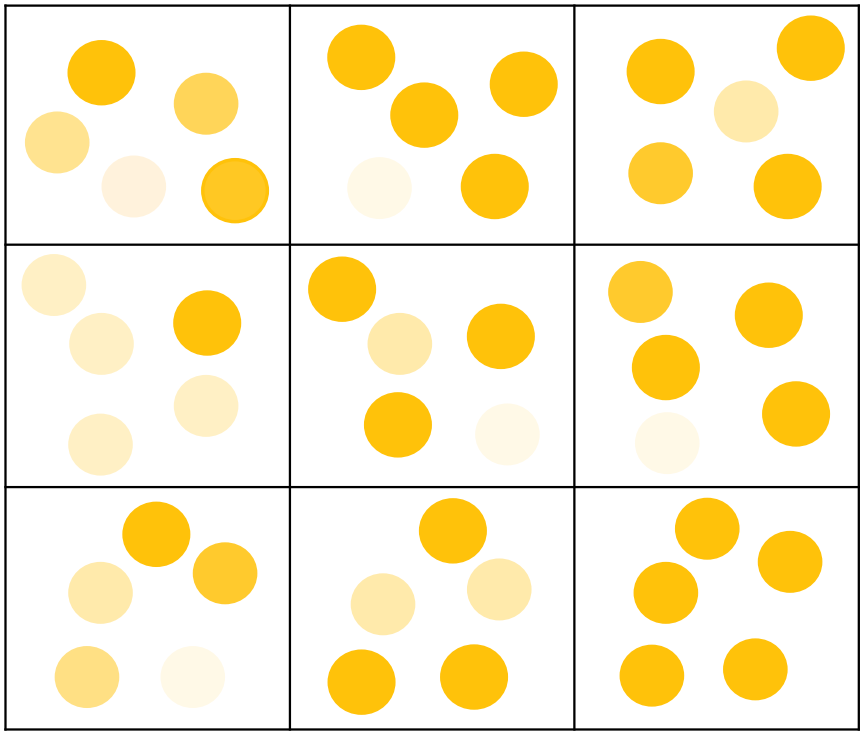


The problem of continuous variables

Categorical variable-based segregation

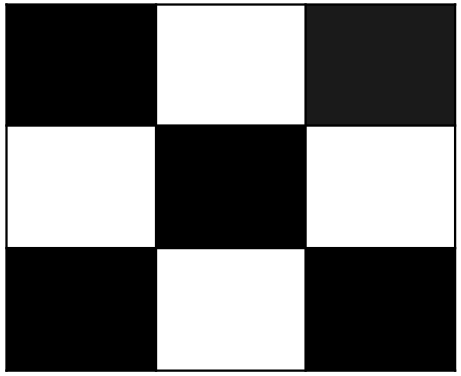


Continuous variable-based segregation



The problem of space

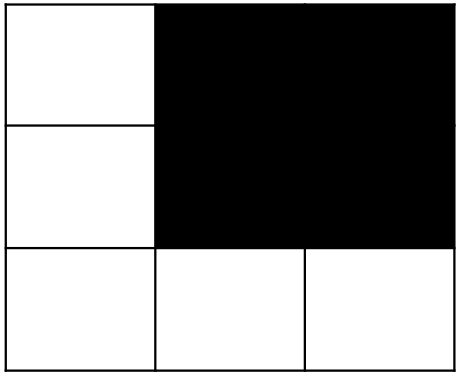
City A

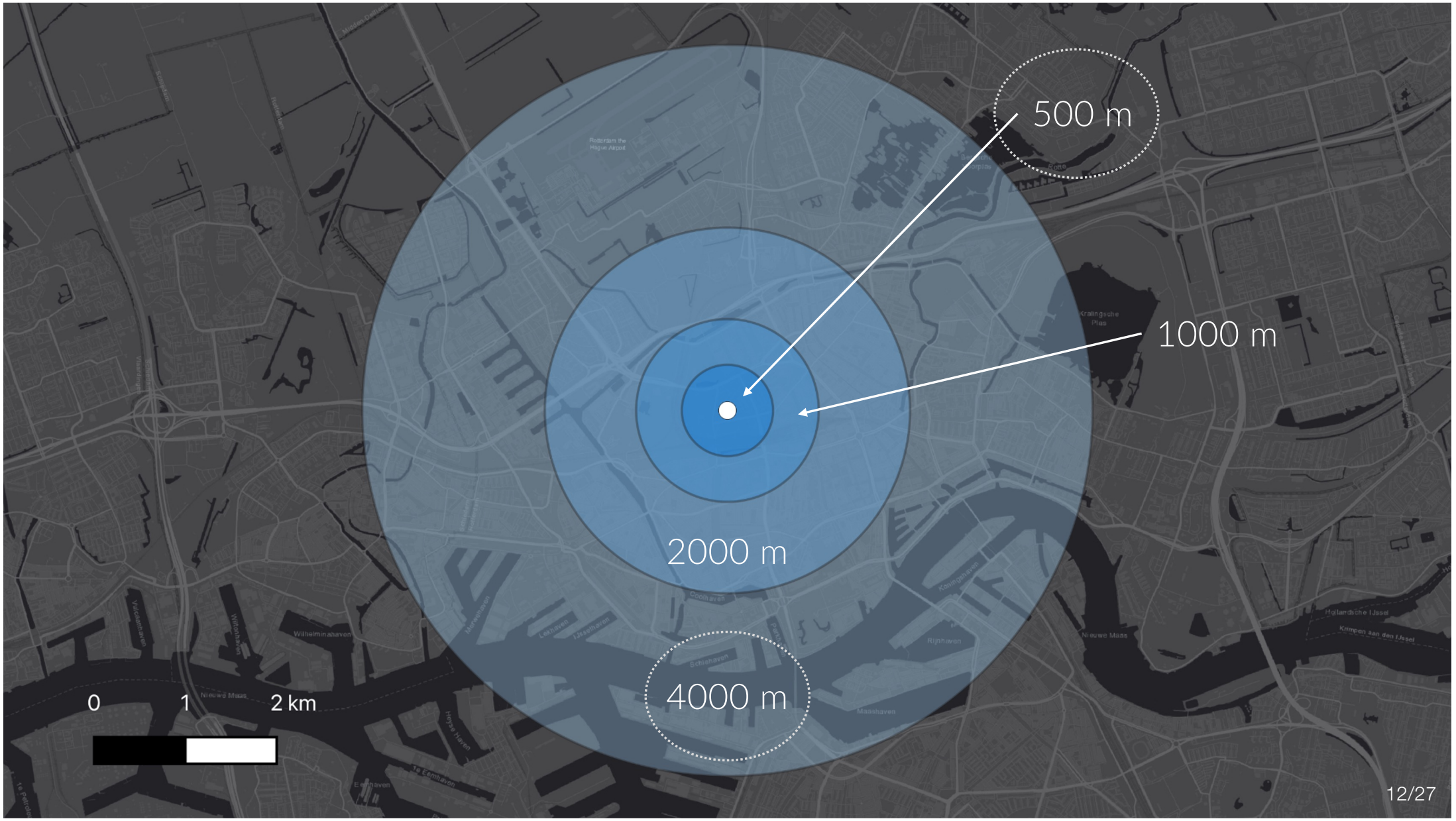


Are these two cities equally segregated?

(White, 1983)

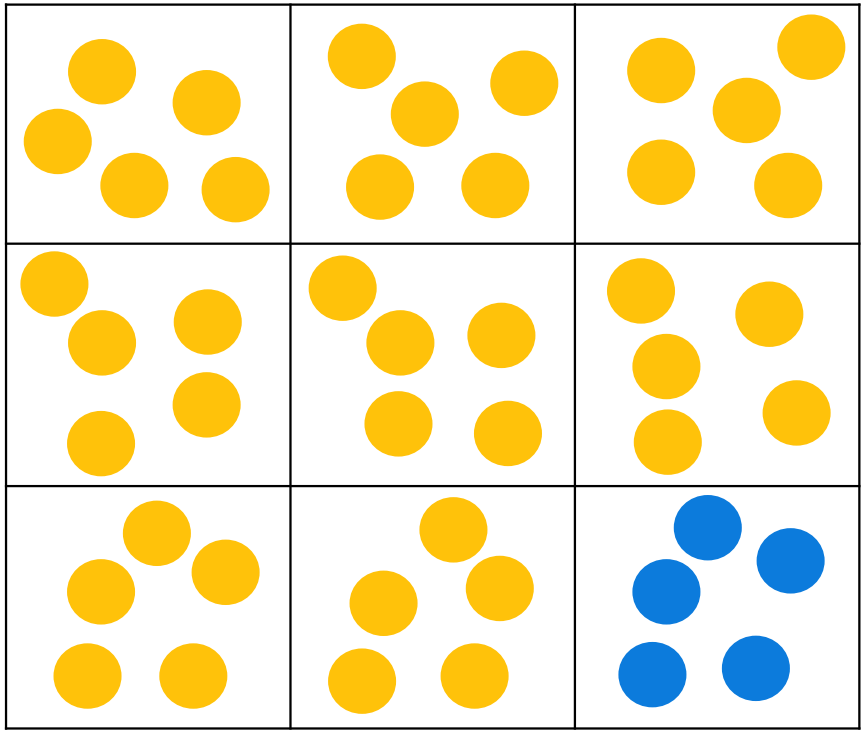
City B




500 m
1000 m
2000 m
4000 m
0
1
2 km

The problem of population structure

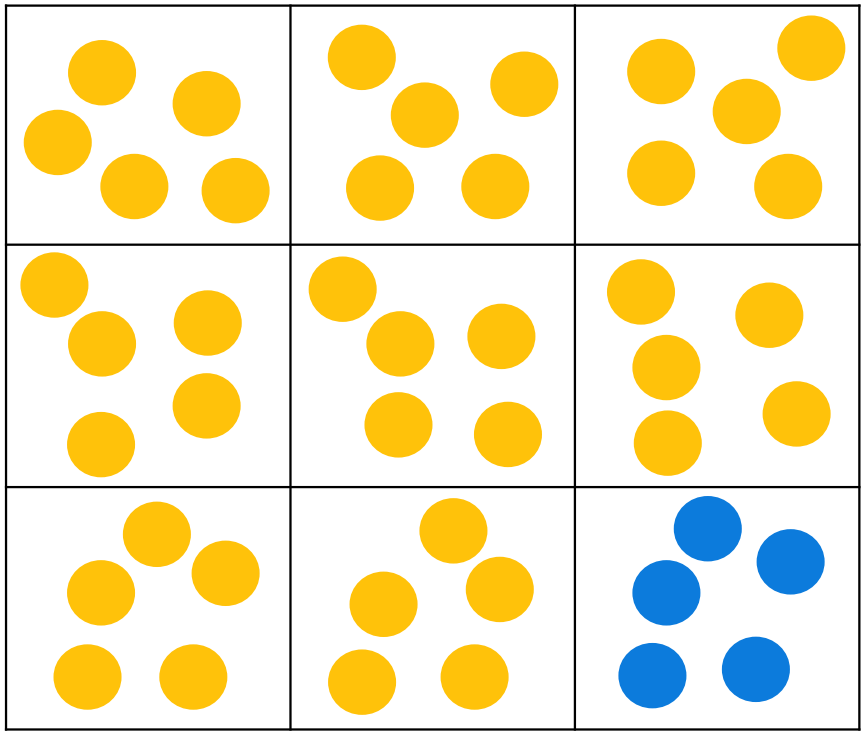
Group A: 40 people; Group B: 5 people



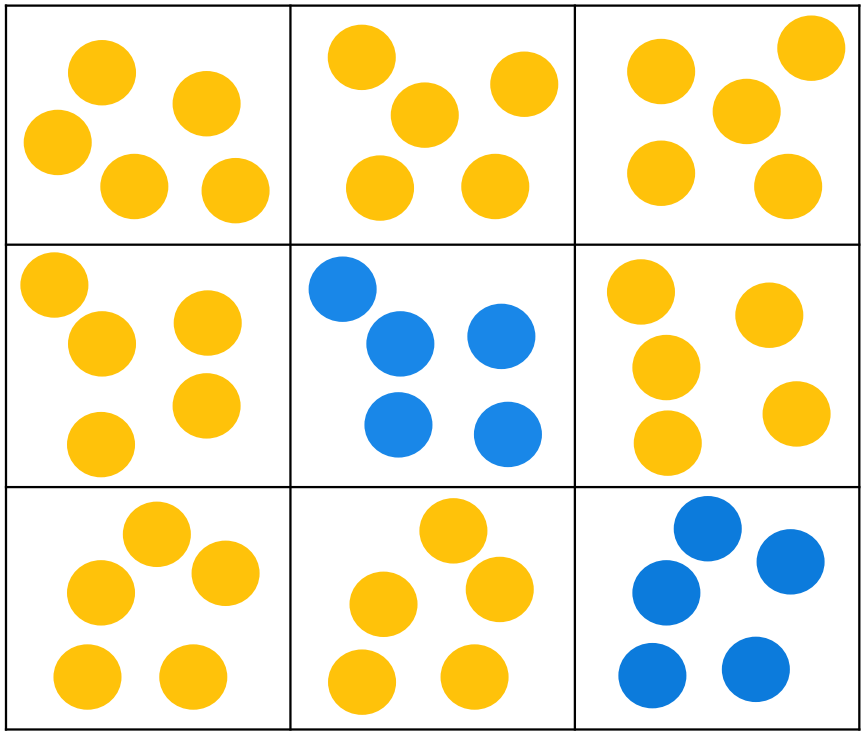
The problem of population structure

Are these two cities equally segregated?

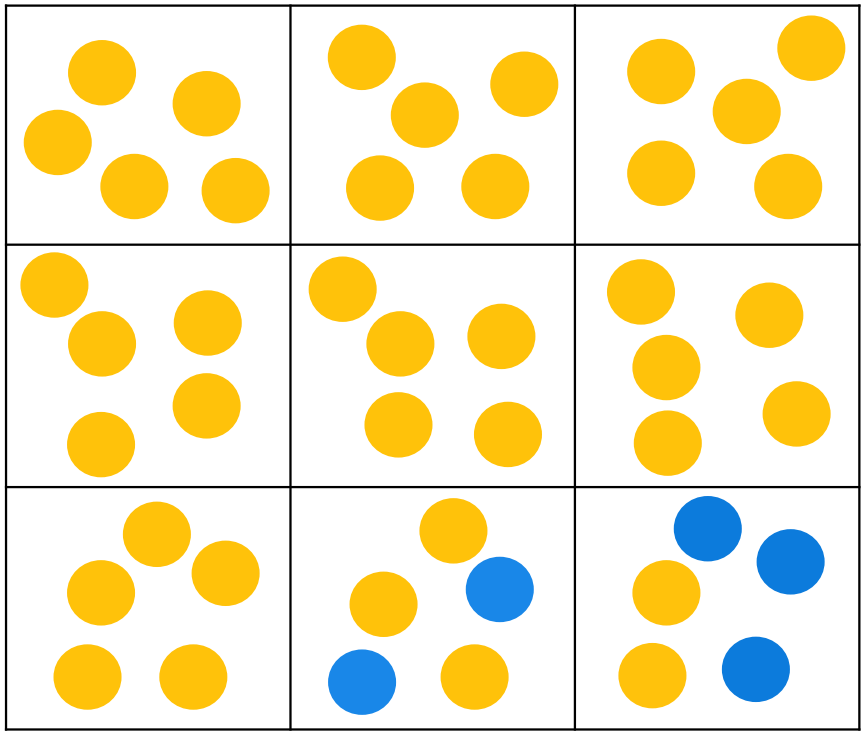
Group A: 40 people; Group B: 5 people



Group A: 35 people; Group B: 10 people

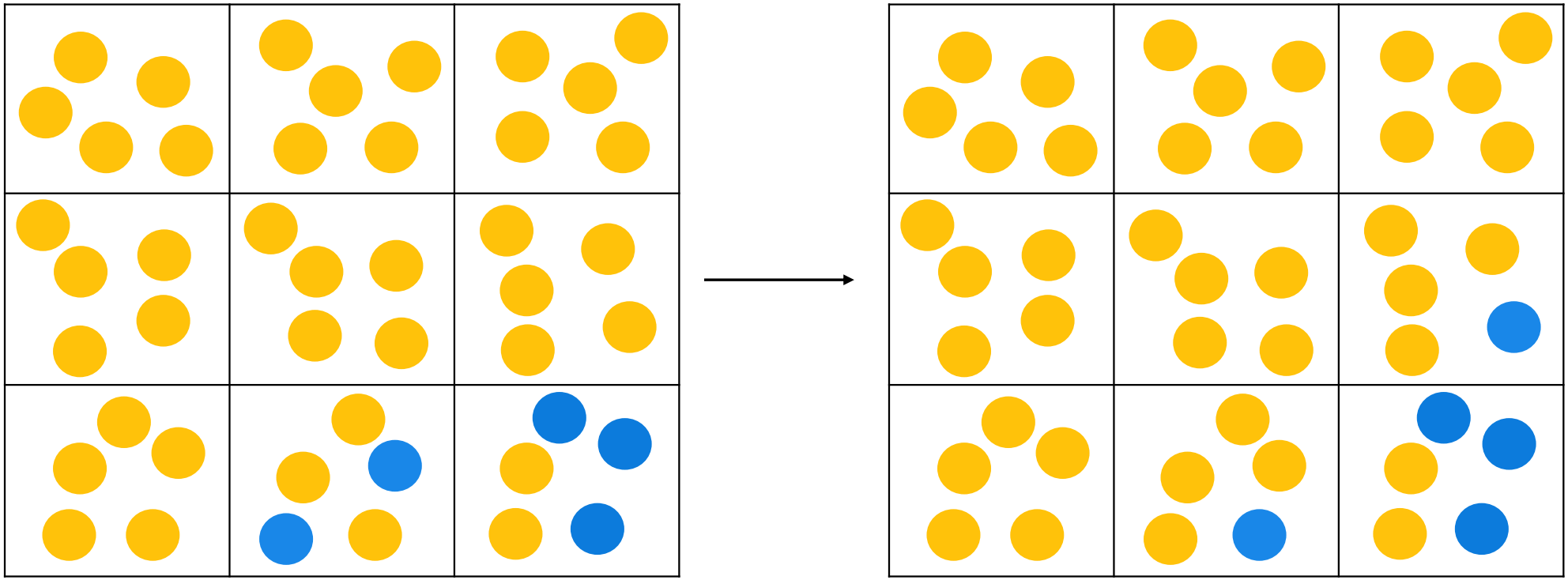


The problem of population structure



The problem of population structure

Small compositional differences can produce large differences in segregation estimates



See: Carrington & Troske (1997)l; Reardon, Bischoff, Owens, & Townsend (2018); Logan, Foster, Xu, & Zhang (2020); Leung-Gagné & Reardon (2023)

The three problems

The continuous nature of some forms segregation → *No straightforward operationalization*

Space
- *Segregation measurements depend a lot on the geographic unit of analysis*
- *Segregation can happen in large-scale patterns (i.e. district-wide segregation) or in a small-scale pattern (i.e. mosaic-like segregation)*

Population structure
- *Some indices depend on relative group sizes by design*
- *Small counts can produce unstable estimates and substantial unevenness by chance*

Indexes

R packages for segregation

Gaps